

17 For double-slit, fringe separation x is given by

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$$x = \frac{\lambda D}{a}$$

$$3.0 \times 10^{-3} = \frac{(600 \times 10^{-9})D}{a}$$

$$x_2 = \frac{(300 \times 10^{-9})D}{0.5a} = x = 3.00 \times 10^{-3} \text{ m}$$

Answer: C

18 q and $4q$ will repel each other. Thus, for the net force on q to be zero,